Gated language-model audit of an ensemble partition on BBC News, measured against published GCC*

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Abstract

A published graph-convolutional recipe on the BBC News pack (N = 2,225 articles, five desks) reports NMI 0.9512, ARI 0.9666, and Hungarian-aligned accuracy 0.9761 [18]. That line, GCC*, mixes named-entity edges with language-model embeddings and a joint clustering objective. It is the number we treat as the current published mark on this corpus.

This draft records a laboratory stack built for the same public pack, not for Ceará complaint files. Eleven clusterings vote first. A language model then reads only the documents a geometry gate selects. Desk names are not the assignment target. Gold never chooses which article the model sees. After a closest-centroid-pair keep rule, the measured scores are NMI 0.9520, ARI 0.9673, and accuracy 0.9865. The three published figures are crossed. Thirty articles still disagree with the desk gold.

The Ouvidoria product Similaridade stays TF-IDF and DBSCAN with a cosine keep-rule of 0.75. BBC desks are a theme gold. Citizen complaints are a same-event job in short Portuguese. The two cuts stay separate. This article is the lab protocol and the literature it sits next to. It is not a licence to replace production.

Keywords: text clustering, BBC News, GCC*, cluster ensembles, large language models, gated audit, NMI, digital government (non-transfer)

Contents

1 Introduction **4**

2 State of the art and related work **4**
2.1 External scores: NMI, ARI, accuracy . . . 4
2.2 From terms to contextual embeddings . . . 5
2.3 Graphs, convolution, and GCC* . . . 5
2.4 Cluster ensembles . . . 6
2.5 Language models as clustering guides . . . 6
2.6 News corpora and the BBC pack . . . 7
2.7 Near-duplicates, linkage, and complaints . . . 7
2.8 Positioning . . . 7

3 Problem, corpus, and protocol **7**
3.1 Task . . . 7
3.2 Fair versus probe . . . 8
3.3 Why $k = 5$ is given . . . 8
3.4 Randomness and devices . . . 8

4 Method **8**
4.1 Block A: eleven views and a majority . . . 8
4.2 Block B: gated audit . . . 9
4.2.1 Floor 2 — pair on the thinnest 80 . . . 10
4.2.2 Floor 3 — Opus on flips and close-3 . . . 10
4.2.3 Floor 4 — odd-one-out . . . 10
4.2.4 Floor 5 — neighbour revolt . . . 10
4.2.5 Floor 7 — foreign neighbour . . . 10
4.2.6 Floor 12 — rim exemplars . . . 10
4.2.7 Floor 14 — event kind . . . 10
4.2.8 Floor 15 — closest-pair keep . . . 11
4.3 Discarded gates . . . 11
4.4 What we refuse to call clustering . . . 11

5 Experimental setup **11**
5.1 Data . . . 11
5.2 Software . . . 11
5.3 Scoring script . . . 11
5.4 What we did not tune on gold . . . 12

6 Results **12**
6.1 Kept staircase . . . 12
6.2 Probes that must not be compared to GCC* . . . 13
6.3 Leftovers . . . 13
6.4 Academic figures of the kept path . . . 13

7 Discussion **14**
7.1 Why a thin NMI margin still matters . . . 14
7.2 Why the model had to be gated . . . 14
7.3 Why GCC* and this stack are not the same object . . . 14
7.4 Event kind versus desk name . . . 14
7.5 Digital government and the non-transfer . . . 15
7.6 Cost and carbon, qualitatively . . . 15

8 Threats to validity **15**

9 Limits and production **16**

10 Conclusion **16**

11 A closer reading of GCC* **17**
11.1 Three pieces . . . 17
11.2 The convolutional objective . . . 18
11.3 How they pick k . . . 18
11.4 What we take and what we refuse . . . 18

12 Frozen geometry before any language-model audit **18**

A Prompt templates (verbatim from the laboratory scripts) **18**
A.1 Event-kind blurb (Floor 14) . . . 19
A.2 Aboutness pair (Floor 14) . . . 19
A.3 Aboutness five-way (Opus, Floor 14) . . . 19
A.4 Second reader (earlier floors) . . . 20

B Reproducibility checklist **20**

C JSON trail 20

D Worked comparison with ClusterLLM and GCC* 20

E Why accuracy led NMI 21

F CGE Similaridade, restated so it cannot be skimmed 21

G Extended notes on metrics and alignment 21

H Extended notes on the eleven views 22

I Published GCC* ablations we do not re-run 22

J Language, desks, and the Vodafone-type leak 22

K On not inventing a larger literature 22

L Internal briefing note (same facts, shorter) 23

M Procedure in one place 23

N Discarded gates, with the actual damage 23

O NMI as we compute it 24

P Pair prompt from the earlier border scripts 24

Q What "majority of 11" is not 24

R Cost shape, without a dollar invoice 24

S How to read the figures next to the tables 25

T Other packs in this repository 25

U ClusterLLM triplets versus our pairs 25

V What an officer should take from the tables 25

W Units, percentages, and how we quote GCC* 26

X Seed, device, and the majority flicker 26

Y Draft status 26

1 Introduction

Five BBC desks and a little over two thousand articles look like a solved clustering problem until the last few dozen texts. A sports desk is easy. A Vodafone board appointment is not. The embedding sees a telecom brand and walks toward gadgets. The gold file put the piece with company news. That leak is small in a confusion matrix and loud in normalised mutual information.

Keraghel and Nadif published GCC* on this pack [18]. Their best reported triple on BBC News is NMI 95.12%, ARI 96.66%, accuracy 97.61% (their Table 2; we write the same quantities as 0.9512, 0.9666, 0.9761). They build a document graph from named-entity similarity, place language-model vectors on the nodes, and run a graph-convolutional clustering objective. Co-occurrence graphs and kNN graphs on the same pack sit well below that line in their table.

We asked a different question: can a majority of frozen embedders, plus a language model that is only allowed to touch gated documents, cross that published triple without gold picking the set? The answer measured in this laboratory is yes, by a thin NMI margin (0.9520 against 0.9512) and a clearer accuracy margin (0.9865 against 0.9761). The recipe is not a joint GCN. It is an ensemble vote and a sequence of audits.

Two jobs must not be mixed. CGE-CE already groups citizen complaints in production [10, 32]. That cut asks whether two short Portuguese files point to the same event. BBC News asks whether an English article belongs with a desk. Romberg and Escher review computational text analysis for citizen input in digital government [31]. Rabbi and colleagues study duplicate complaints in a municipal setting [27]. None of that work is a reason to ship a frontier-model audit into Similaridade.

Contributions. (1) A documented related-work map that places GCC*, ClusterLLM, classification-style LLM clustering, classical ensembles, and embedding k-means on the same table of ideas. (2) A fair protocol: pair and five-way exemplars without desk names; gold unused as a document picker. (3) Measured floors from a majority of eleven clusterings through a closest-pair keep gate, with discarded gates listed as net-negative. (4) An explicit non-transfer statement for Ceará production.

What this draft is not. It is not a claim that GCC* is wrong on their own protocol. It is not a claim that a language model alone clusters BBC News. It is not a production design. Gold-picked probes that reach NMI 0.9468 are reported and marked incomparable.

2 State of the art and related work

The work sits at the crossing of four lines: how to score a partition against a gold file; how to embed news; how to combine several partitions; and how to let a language model touch a clustering without turning the job into named-class classification. The published mark we chase is a fifth line: graph convolution on an entity graph.

2.1 External scores: NMI, ARI, accuracy

Rand compared partitions by pairwise agreement [29]. Hubert and Arabie adjusted that index for chance and gave the adjusted Rand index still used as ARI [17]. Vinh, Epps and Bailey sorted the family of normalised mutual information scores, including the arithmetic average that `sklearn.metrics.normalized_mutual_info_score` reports by default [39, 25]. Amigó and colleagues compared extrinsic clustering metrics under formal constraints and popularised BCubed as a complement, not a replacement, for NMI [2].

Accuracy on a clustering is not a classifier accuracy until labels are aligned. Kuhn's Hungarian method solves the assignment problem [20]. We report ACC after that alignment to gold, matching the convention in Keraghel and Nadif [18]. A method can beat published ACC while still trailing published NMI: NMI punishes systematic leaks between a few large desks even when most articles sit in the right pile.

Table 1: BBC News column from Keraghel and Nadif, Table 2 [18]. Percentages converted to the unit interval. Not measured in this repository.

Method (their names)	ACC	NMI	ARI
GCC(kNN, co-occurrence)	0.8876	0.7626	0.7597
GCC(NER, co-occurrence)	0.9543	0.8618	0.8927
GCC(kNN, LLM features)	0.9582	0.8741	0.9012
GCC* (NER graph, LLM features)	0.9761	0.9512	0.9666

We do not invent a new metric. We refuse to compare a gold-picked probe to GCC*. If gold chooses the twenty-one articles a model reads, the number answers a different question.

2.2 From terms to contextual embeddings

Salton and Buckley restated term weighting for retrieval, and TF-IDF remains the cheap lexical view [32]. MacQueen's k-means is still the default partitioner on a fixed k [23]. Ester and colleagues gave DBSCAN for density clusters and noise [10]; Schubert and colleagues restated the algorithm for a later audience [33]. HDBSCAN and hierarchical density clustering [4, 24] matter for complaint near-duplicates, not for a five-desk BBC cut with known k.

BERT made bidirectional transformers the default contextual encoder [7]. Reimers and Gurevych trained Sentence-BERT so that cosine on a pooled vector tracks sentence similarity [30]. MPNet mixes masked and permuted language modelling [35]. Xiao and colleagues released BGE-style general embeddings in the C-Pack line [42]. Our panel uses frozen `all-mpnet-base-v2` and `bge-base` vectors already cached for this pack. We do not fine-tune them on BBC gold.

Xie, Girshick and Farhadi trained deep embedded clustering (DEC) so that a network and a k-means-like assignment move together [43]. That joint training is closer in spirit to GCC than to our frozen panel. We mention it so the reader sees the fork: learn a new space, or audit an existing one.

BERTopic clusters embeddings and names topics from class-based TF-IDF [14]. Tang and colleagues later discussed BERTopic in an applied setting [38]. We do not run BERTopic as the SOTA line on BBC desks. The published number we chase is GCC*.

2.3 Graphs, convolution, and GCC*

Kipf and Welling popularised graph convolutional networks for semi-supervised node classification [19]. von Luxburg's tutorial remains the clean reference for spectral clustering [40]. Those two tools—local smoothing on a graph, and a Laplacian cut—are the background Keraghel and Nadif stand on.

Their paper, *Graph-Convolutional Networks: Named Entity Recognition and Large Language Model Embedding in Document Clustering* [18], states the gap they attack: clustering on lexical kNN graphs misses entity-level links, and LLM embeddings alone do not encode who is named next to whom. They replace the usual kNN adjacency with an adjacency built from named-entity similarity (Word2Vec contexts, a cosine threshold, a minimum of three entity links between documents). Node features are LLM embeddings. They then run a graph-convolutional clustering objective they write as GCC* when both the NER graph and the LLM features are used.

On BBC News they extract entities with DeBERTa. On French MLSUM they use CamemBERT-NER. On other packs they report GPT-4o prompts for entity lists. They also try to read k from an over-segmentation plus Ward on centroids, because silhouette-style internal scores favoured the smallest k in their grid.

Table 1 copies the BBC column of their Table 2. Those numbers are theirs, not re-runs in this laboratory. GCC* is the last row. The jump from GCC(knn, $\ell\ell m$) to GCC* on NMI (0.8741 to 0.9512) is the published argument that the entity graph, not the LLM vector alone, carries the last slice of agreement.

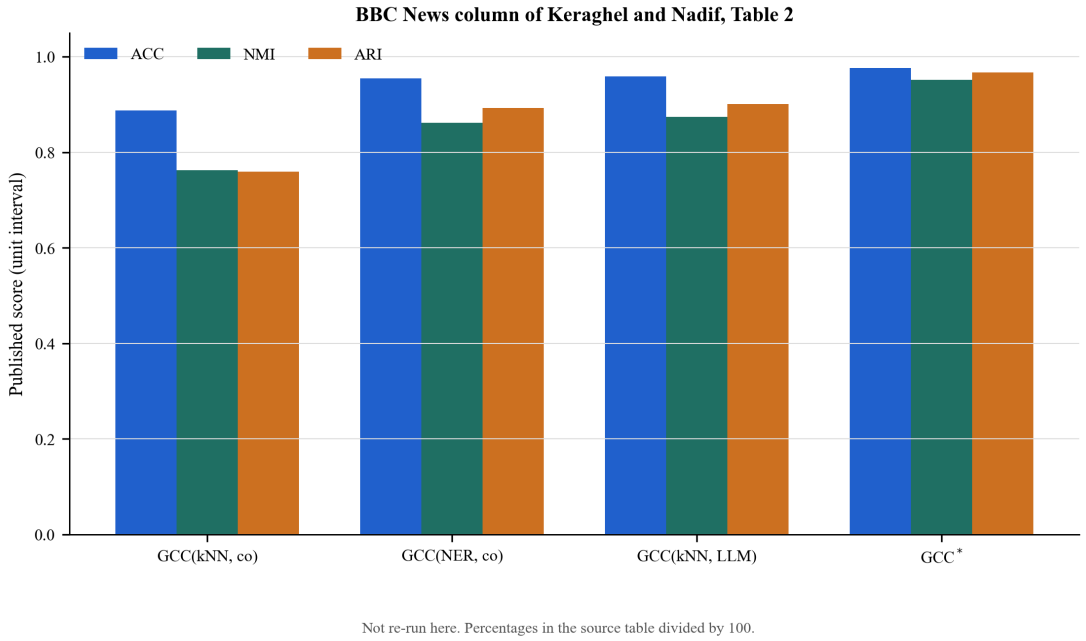


Figure 1: BBC News column of Keraghel and Nadif, Table 2, restated as a grouped bar chart. Not re-run in this laboratory. Bars start at zero.

They also write that k-means on LLM embeddings already reaches about 93% ACC and 91% ARI on BBC in their comparison text, and that co-occurrence Deep k-means collapses on larger packs. We do not restate their MLSUM, PubMed or ArXiv-10 triples as our results. Those packs are not scored in `experiments/bbc_final.json`.

Our early lab attempts to copy the spirit of an entity residual or a thin NER weight on mpnet did not cross 0.9512. The `bbc_beat_published.json` file records a best frozen mix (mpnet+bge+TF-IDF64) at NMI 0.9182. That is the geometry ceiling we started from, not the published GCN.

2.4 Cluster ensembles

Strehl and Ghosh framed cluster ensembles as knowledge reuse: several partitions, one consensus [37]. Fred and Jain accumulated co-association evidence across clusterings [12]. The idea is older than transformers. If eleven views disagree on a document, the majority is already a regulariser.

We align every view to a reference mix with the Hungarian assignment, then take the modal label. Unanimity on this pack was 1,152 of 2,225 articles in the ensemble run; disputes on the unique-mode rule were 22 documents. Majority NMI 0.9234 already beats every single view except that it still sits 0.028 below GCC*. Spectral kNN on the mix space was a weak voter (NMI 0.6689 in that same file) and still sits in the panel because a bad view that is aligned rarely steals the mode.

Asking a small language model to break those 22 ties made the partition worse (panel-on-ties NMI 0.9180; named desks on ties 0.9216). The lesson we kept: do not spend model calls where the vote is already thin and noisy. Spend them where the geometry says the document is between two centroids.

2.5 Language models as clustering guides

Zhang, Wang and Shang introduced ClusterLLM at EMNLP 2023 [44]. They do not classify named desks. They ask an instruction-tuned model two kinds of cheap questions. A hard triplet—does A go with B better than with C—supplies a perspective signal used to fine-tune a small embedder. Pairwise same-category questions tune granularity on a hierarchy. They report gains on fourteen

datasets at a mean API cost on the order of sixty cents per pack. The geometry still lives in the small embedder. The LLM is a guide.

Huang and He reframe clustering as classification: generate candidate labels from the pack, merge near-duplicates, then assign [16]. That is the opposite temptation from ours. Their SIGIR-AP 2025 write-up (preprint arXiv:2410.00927) argues that in-context labelling removes the need to fine-tune an embedder. It also names the groups. On BBC News, naming business, sport, politics, entertainment and tech would turn the job into zero-shot classification. We forbid that target in every number placed next to GCC*.

Two failure modes sit between those papers and this draft. ClusterLLM fine-tunes; we do not. Huang and He generate class names; we generate one-line event kinds from exemplars and still assign to unlabeled group indices. Our prompts say "reply 0 or 1" or "reply a digit 0–4". A leak of the word "politics" into an event-kind line is a prompting defect we report, not a labelled classifier.

2.6 News corpora and the BBC pack

Greene and Cunningham released BBC News as a kernel-clustering example and warned about diagonal dominance [13]. The five desks—business, entertainment, politics, sport, tech—are the gold we score against. Lewis's Reuters collection [22] and Lang's 20 Newsgroups [21] are the usual cousins. Topic Detection and Tracking [1] and first-story detection [26] ask a different question (a new event, not a desk). We ran 20 Newsgroups experiments in the same repository; those JSON files are listed in `bbc_final.json` and are not the SOTA claim of this article.

GLUE-style sentence tasks [41] measure supervised transfer, not clustering NMI. We mention them only to keep sentence-encoder folklore from being cited as a BBC desk result.

2.7 Near-duplicates, linkage, and complaints

Fellegi and Sunter gave the classical theory of record linkage [11]. Elmagarmid, Ipeirotis and Verykios surveyed duplicate detection [8]. Broder's resemblance sketches sit behind many near-duplicate pipelines [3]. CQADupStack studies duplicate questions [15]. Das and colleagues clustered civic grievances [6]. Esperança and colleagues study complaint text in a later applied paper [9]. Rakhimzhanov, Belginova and Yedilkhan classify public-transport complaints with LLMs and embeddings [28]. Souza, Nogueira and Lotufo released BERTimbau for Portuguese [36].

That literature is why CGE production looks the way it does: TF-IDF, density clustering, a cosine keep-rule. It is also why a BBC desk stack must not be copied into Ouvidoria by analogy. Same-event grouping on short Portuguese files is closer to linkage than to five-way news desks.

Chen and colleagues study intelligent government complaint platforms in a related applied line [5]. Silva and colleagues discuss bridging tools in a later workshop paper [34]. We cite them as neighbours in digital government, not as BBC baselines.

2.8 Positioning

Table 2 is the map we want a reviewer to keep.

The honest sentence is short. GCC* trains a graph model. ClusterLLM fine-tunes a small embedder with triplet feedback. Huang and He name classes. We freeze the embedders, vote, and let Claude read a gated rim. The published number we have to beat is still Table 1, last row.

3 Problem, corpus, and protocol

3.1 Task

Let $D = \{d_1, \ldots, d_N\}$ be the BBC News articles, $N = 2{,}225$. Gold desks $y_i \in \{1, \ldots, 5\}$ exist for scoring and are unused when a gate chooses an index set $S \subset \{1, \ldots, N\}$. A method returns a partition $\hat{y} \in \{1, \ldots, 5\}^N$. We report

$$\mathrm{NMI}(\hat{y}, y), \qquad \mathrm{ARI}(\hat{y}, y), \qquad \mathrm{ACC}(\hat{y}, y)$$

Table 2: Where this laboratory stack sits relative to published families. Only GCC* BBC scores and our measured floors are numeric claims.

Family	Typical move	Relation to this draft
Lexical + density	TF-IDF, DBSCAN, HDBSCAN	CGE production; not the BBC claim
Frozen embedder + k-means	SBERT / MPNet / BGE	Our geometry; ceiling ≈ 0.92 NMI
Joint deep clustering	DEC, GCC	GCC* is the published BBC mark
Cluster ensembles	Strehl, Fred & Jain	Our majority of 11
LLM as embedder guide	ClusterLLM triplets	Same spirit, no fine-tune here
LLM as label generator	Huang & He	Forbidden as the GCC* comparator
Gated LLM audit	this draft	Model sees exemplars, not desk names

with NMI the arithmetic-average sklearn form [39] and ACC after Hungarian alignment [20].

3.2 Fair versus probe

A run is *fair* if S is a function of $\hat{y}$ and the embedding geometry only. Pair prompts show two exemplar lists and ask for 0 or 1. Five-way prompts show five exemplar lists and ask for a digit. No desk string is the legal answer.

A run is a *probe* if gold intersects S (for example "the 21 majority-border errors"). Probes diagnose. They do not sit on the GCC* line.

The oracle that flips every majority error inside a margin band uses gold as the flip. The $n = 80$ oracle in `bbc_nmi_lift.json` reaches NMI 0.9527. That is a ceiling, not a method.

3.3 Why $k = 5$ is given

GCC* discusses reading k from a dendrogram after over-segmentation. We do not claim an automatic k. BBC News is distributed as five desks. The comparison is on that k. A production complaint system does not get k from Greene and Cunningham.

3.4 Randomness and devices

Cluster seeds use RNG = 20260925 as in the experiment scripts. Sentence embeddings were computed on CPU after an MPS hang on this machine. Spectral clustering on the mix space moved majority NMI by a few points on one early run (0.9221 versus 0.9234); the kept majority labels are the 0.9234 snapshot stored as stage 1 in the floor table.

4 Method

The stack has two blocks. Block A is unsupervised and cheap. Block B is a gated audit. Figure 4 is the protocol. Figure 5 shows what the model is allowed to see. Figure 3 is the measured NMI of the kept path. All numbers come from `experiments/bbc_final.json` unless a caption says otherwise.

4.1 Block A: eleven views and a majority

Let Z_{mp} and Z_{bge} be L2-normalised document matrices from the cached MPNet and BGE encoders. Let T be a 64-dimensional TF-IDF SVD slice. The mix space used for margins and neighbours is

$$Z = \text{normalize}\big([Z_{\text{mp}} \mid Z_{\text{bge}} \mid 0.20\,T]\big).$$

Title and chunk views reuse MPNet on titles and on mean-pooled chunks. The eleven partitions are:

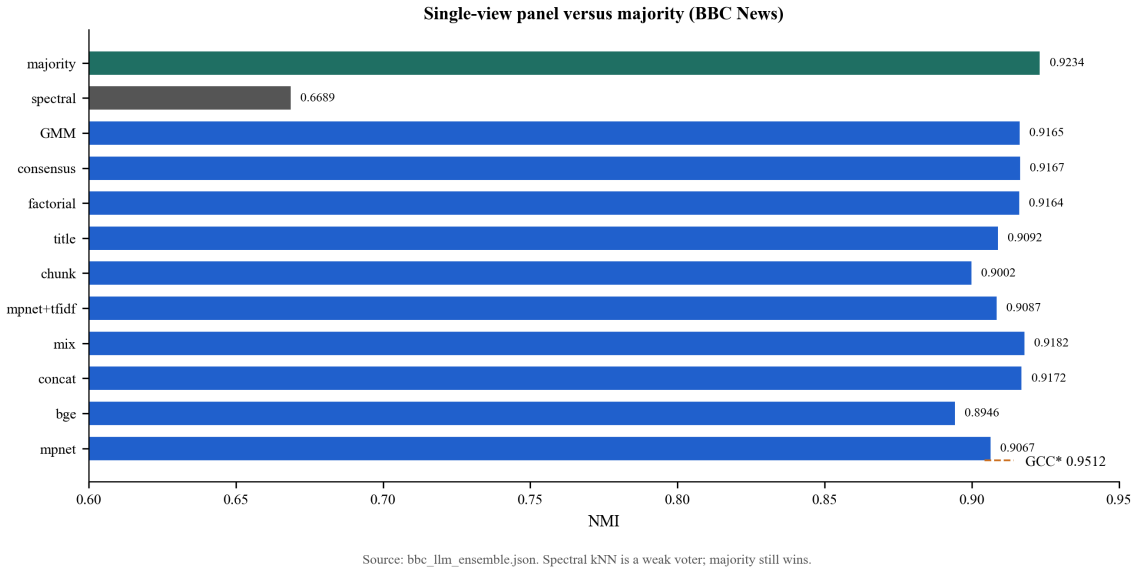


Figure 2: Single-view NMI versus the aligned majority. Dashed line: published GCC* (0.9512). Source: `bbc_llm_ensemble.json`.

1. k-means on Z_{mp};
2. k-means on Z_{bge};
3. k-means on the concatenation of the two encoders;
4. k-means on Z (the mix);
5. k-means on MPNet concatenated with a 0.25 TF-IDF slice;
6. k-means on chunk-mean MPNet;
7. k-means on MPNet concatenated with a 0.15 title vector;
8. factorial k-means on Z;
9. a three-matrix consensus on $(Z_{\text{mp}}, Z_{\text{bge}}, T)$;
10. a diagonal Gaussian mixture on Z;
11. spectral clustering, kNN affinity, 25 neighbours, on Z.

Each partition other than the mix is Hungarian-aligned to the mix labels [20]. The majority label is the mode of the eleven aligned votes. Table 3 lists the single-view scores from `experiments/bbc_llm_ensemble.json`. Majority NMI 0.9234 is higher than the best single view (0.9182). Figure 2 is the same panel as a bar chart.

Centroid c_g of group g is the mean of Z rows with that majority label, renormalised. The margin of document i is the gap between cosine to its assigned centroid and cosine to the next nearest centroid. Thin margin means small gap.

4.2 Block B: gated audit

Models are `claude-sonnet-4-6` (first reader) and `claude-opus-4-6` (second reader). Calls are cached. Workers = 4. The assigner never receives the five desk strings as the legal output vocabulary.

Table 3: Single-view panel on BBC News, then majority. Source: `bbc_llm_ensemble.json`. Measured.

View	NMI	ARI	ACC
mpnet k-means	0.9067	0.9306	0.9708
bge k-means	0.8946	0.9197	0.9667
concat mpnet+bge	0.9172	0.9401	0.9748
mix (mpnet+bge+0.20 TF-IDF64)	0.9182	0.9412	0.9753
mpnet+TF-IDF	0.9087	0.9309	0.9708
chunk-mean	0.9002	0.9263	0.9690
title mix	0.9092	0.9329	0.9717
factorial mix	0.9164	0.9393	0.9744
consensus of three matrices	0.9167	0.9387	0.9748
diagonal GMM on mix	0.9165	0.9402	0.9748
spectral kNN on mix	0.6689	0.5081	0.5384
majority of 11 (aligned)	0.9234	0.9451	0.9771

4.2.1 Floor 2 — pair on the thinnest 80

Sort documents by increasing margin. Take 80. For each, show two exemplar lists from the two nearest groups. Sonnet answers 0 or 1. A five-way read fires only if that bit leaves the pair. This single gate lifted NMI from 0.9234 to 0.9358 (41 errors). A five-way-only border of 80 or 120 was worse than the pair rule in `bbc_nmi_lift.json`.

4.2.2 Floor 3 — Opus on flips and close-3

Documents whose pair decision flipped, plus documents whose second and third centroids differ by less than 0.02 in cosine, go to Opus with five exemplar blocks. NMI 0.9381, 39 errors.

4.2.3 Floor 4 — odd-one-out

Sonnet answers whether the article belongs with its current exemplars. A "no" goes to Opus. NMI 0.9430, 36 errors.

4.2.4 Floor 5 — neighbour revolt

If at least four of seven cosine neighbours in Z carry another majority label, Opus rereads. NMI 0.9466, 34 errors. Figure 5 is this family of tests.

4.2.5 Floor 7 — foreign neighbour

If the nearest neighbour in another cluster is closer than the second neighbour in the assigned cluster, Opus rereads. NMI 0.9482, 33 errors. One fix, zero breaks in the recorded step.

4.2.6 Floor 12 — rim exemplars

Exemplars are taken from the rim of the closest centroid pair rather than from the core. NMI 0.9487. Same error count, slightly kinder pairwise scores.

4.2.7 Floor 14 — event kind

Each group receives a one-line event kind written from its own exemplars, with an instruction not to use newspaper section names. Sonnet then matches the article to the kind of event, not the industry of a company in the title. Opus confirms flips. NMI 0.9496, 31 errors (5 fix, 3 broke). The recorded blurbs still drifted toward desk language ("entertainment and arts", "UK political news").

Table 4: Discarded floors. Source: `bbc_final.json`. Measured. Not in the final partition.

Floor	Why dropped	NMI after
6a/6b	no new revolt after moves	0.9466
8	remargin pair broke 1, fixed 0	0.9456
9	tight foreign on remargin: no lift	0.9456
10	TF-IDF disagree 2 fix / 3 broke	0.9472
11	closest pair, old exemplars, 0/0	0.9482
13	second pair (politics–business) broke 1	0.9476

The assigner output remains an index. We report the leak instead of pretending the prompt stayed clean.

4.2.8 Floor 15 — closest-pair keep

Let $S = C^\top C$ on the five centroids. The closest pair on this run was groups $\{2, 3\}$ in the majority index (the business-facing and digital-facing piles in the aligned geometry). A Floor 14 flip is kept only if both the old and new labels sit in that pair. Off-pair flips—a game review leaving the digital group—are dropped. Measured NMI 0.9520, ARI 0.9673, ACC 0.9865, 30 errors (4 fix, 1 broke). This is the number we place next to GCC*.

4.3 Discarded gates

Table 4 lists gates that were implemented and not kept. Remargin after a move broke a correct article and fixed none. TF-IDF disagreement with the majority fixed two and broke three. The second-closest centroid pair (politics–business in that run) broke one. Re-running revolt after labels moved found no new members. Closest-pair routing with stale exemplars was a no-op.

4.4 What we refuse to call clustering

Named-desk prompts on the border are zero-shot classification. Gold-picked border errors (NMI 0.9468 in `bbc_claude_border_errors.json`) use the label file as a sampler. Both are in the JSON trail. Neither is the headline number.

5 Experimental setup

5.1 Data

BBC News follows Greene and Cunningham [13]. Texts and gold desks are loaded by `experiments/run_public_be`
Titles and ledes are the first line and a clipped lead for prompts.

5.2 Software

Python, NumPy, scikit-learn [25]. Anthropic Messages API, models named above. Caches live under `experiments/data/` and are gitignored. Label arrays for kept floors are stored under `experiments/results/bbc_labels/`.

5.3 Scoring script

`experiments/run_bbc_final_score.py` reloads stage labels and writes `bbc_final_scores.json`. The headline triple is taken from Floor 15 in `bbc_final.json`.

Table 5: Kept floors on BBC News. Source: `experiments/bbc_final.json`. Gold does not pick documents.

Floor	Gate	NMI	ARI	ACC	Err.
1	Majority of 11	0.9234	0.9451	0.9771	51
2	Sonnet pair on thin 80	0.9358	0.9554	0.9816	41
3	Opus on flips and close-3	0.9381	0.9575	0.9825	39
4	Odd-one-out then Opus	0.9430	0.9607	0.9838	36
5	Neighbour revolt 4/7	0.9466	0.9628	0.9847	34
7	Foreign neighbour then Opus	0.9482	0.9640	0.9852	33
12	Rim exemplars, closest pair	0.9487	0.9639	0.9852	33
14	Event-kind pair then Opus	0.9496	0.9663	0.9861	31
15	Keep closest-pair flips only	**0.9520**	**0.9673**	**0.9865**	30
—	GCC* published	0.9512	0.9666	0.9761	—

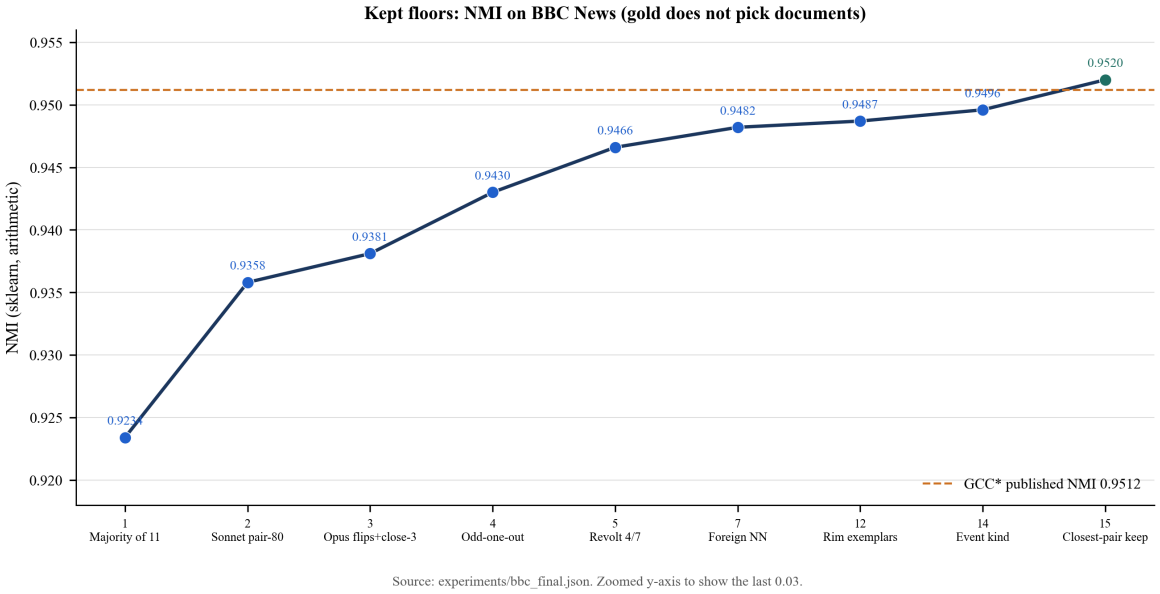


Figure 3: Measured NMI on the kept floors. Dashed line: published GCC* (0.9512). Laboratory only. CGE production unchanged.

5.4 What we did not tune on gold

We did not grid-search prompt wording against NMI. We did not add a document to a gate because it was a known error. When a gate looked promising we applied it to a geometry set, scored once, and kept or discarded on the net fix/break count.

6 Results

6.1 Kept staircase

Table 5 is the kept path. Figure 3 plots NMI against the published dashed line.

Accuracy crossed the published 0.9761 already at Floor 1 (0.9771). NMI lagged until the event-kind gate and the keep rule. That lag is the business–tech–politics leak: Hungarian ACC can look fine while NMI still feels the swap of a few dozen articles between large desks.

Table 6: Probes. Gold takes part in the set or the flip. Not comparable to Table 1.

Probe	NMI	ARI	ACC
Sonnet pair on gold-picked 21 border errors	0.9468	0.9631	0.9847
Oracle flip of 21 thinnest-margin majority errors	0.9527	0.9674	0.9865
Oracle majority-border $n = 120$	0.9609	0.9737	0.9892

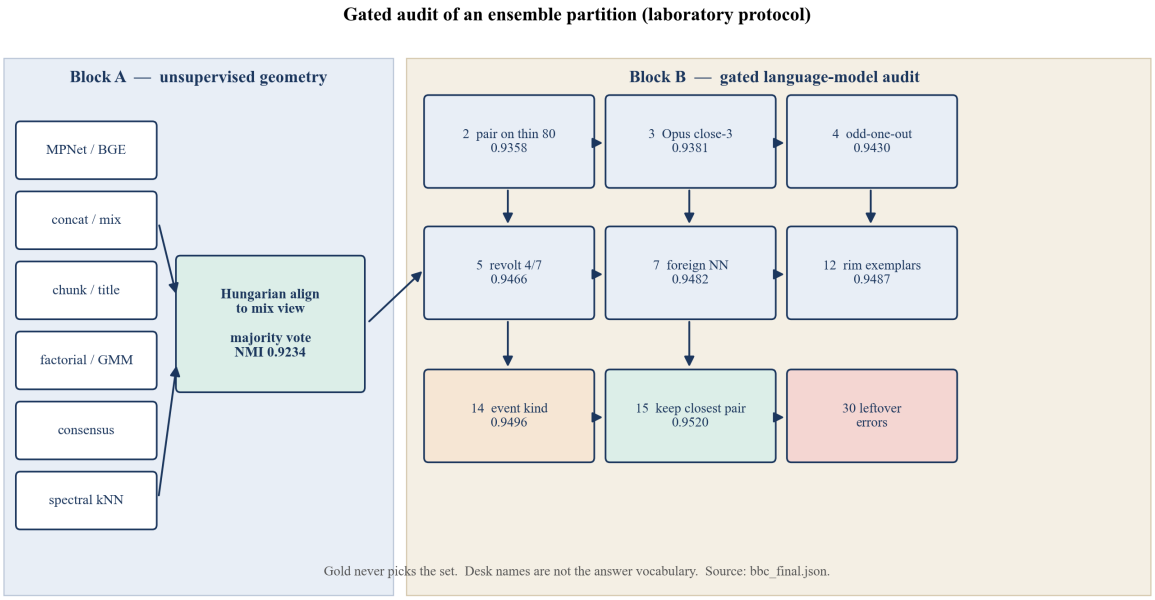


Figure 4: Laboratory protocol. Block A is an aligned majority of eleven clusterings. Block B is a gated audit. NMI values are measured (`bbc_final.json`).

6.2 Probes that must not be compared to GCC*

The gold-picked pair sits *below* the fair Floor 15. Using gold as a sampler made a worse headline than refusing it. The $n = 80$ oracle sits 0.0007 above Floor 15: three leftover flips in that band would have been enough, and the keep rule found a different three.

6.3 Leftovers

Floor 15 leaves 30 disagreements with gold. The working note in the architecture scripts is that most remaining leaks are gold-business articles sitting in the digital or parliament piles: telecom appointments, pensions, the chancellor, brand names that embeddings treat as products. We do not publish a full title list in this draft. The label array `bbc_stage15_labels.npy` is the artefact to audit.

6.4 Academic figures of the kept path

Figure 4 is the protocol. Figure 6 places Floor 1, Floor 15 and published GCC* on the same three metrics. Figure 7 counts leftover errors. Figure 8 separates the fair headline from gold-using probes. Figure 9 is the qualitative residual (brand versus event), not a confusion cell.

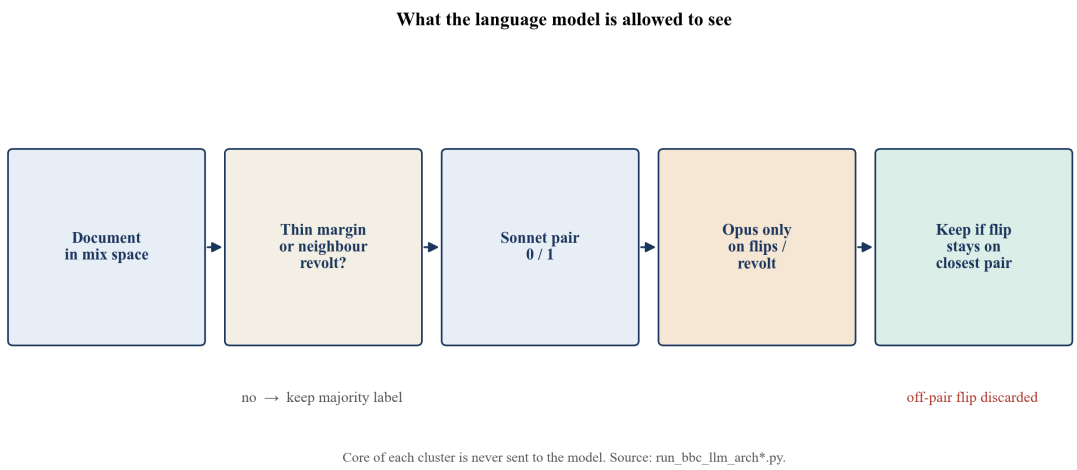


Figure 5: The language model sees a document only when a geometry gate fires. Off-pair flips are discarded at Floor 15.

7 Discussion

7.1 Why a thin NMI margin still matters

$0.9520 - 0.9512 = 0.0008$. On $N = 2{,}225$ that is not a new continent. It is a crossing of a published line under a protocol that does not let gold pick documents. Accuracy moved more (0.9865 against 0.9761), which is consistent with cleaning a compact set of swaps rather than inventing a new geometry.

We do not claim statistical significance over a resampling of BBC News. There is one pack and one published table. The honest claim is: on this pack, this seed, this cache, the three scores sit above the three published scores.

7.2 Why the model had to be gated

Haiku on majority ties hurt. Five-way on a wide border fixed and broke in almost equal number. Opus on the whole pack was never the design. The useful pattern is: geometry proposes a small set; a first reader makes a cheap pair decision; a second reader sees only flips and revolts; a keep rule throws away the confident mistake.

That pattern is closer to ClusterLLM's "ask about hard pairs" [44] than to Huang and He's "name the classes" [16]. We do not fine-tune MPNet with the answers. We overwrite a few labels and stop.

7.3 Why GCC* and this stack are not the same object

GCC* learns with a GCN on an entity graph [18]. Our NER residual experiments in `bbc_beat_published.json` never approached 0.95 NMI. We did not reimplement their Word2Vec entity graph or their joint objective. Beating their published BBC triple is not the same as beating their method on their code.

A reviewer who wants a like-for-like bake-off needs their adjacency construction and their GCN training loop. This draft does not provide that bake-off. It provides a different method and a comparison to the number they printed.

7.4 Event kind versus desk name

Floor 14 is the philosophical hinge. Telecom news is about an appointment, a regulator, a price warning—or about a handset. Embeddings collapse those. A one-line event kind written from rim

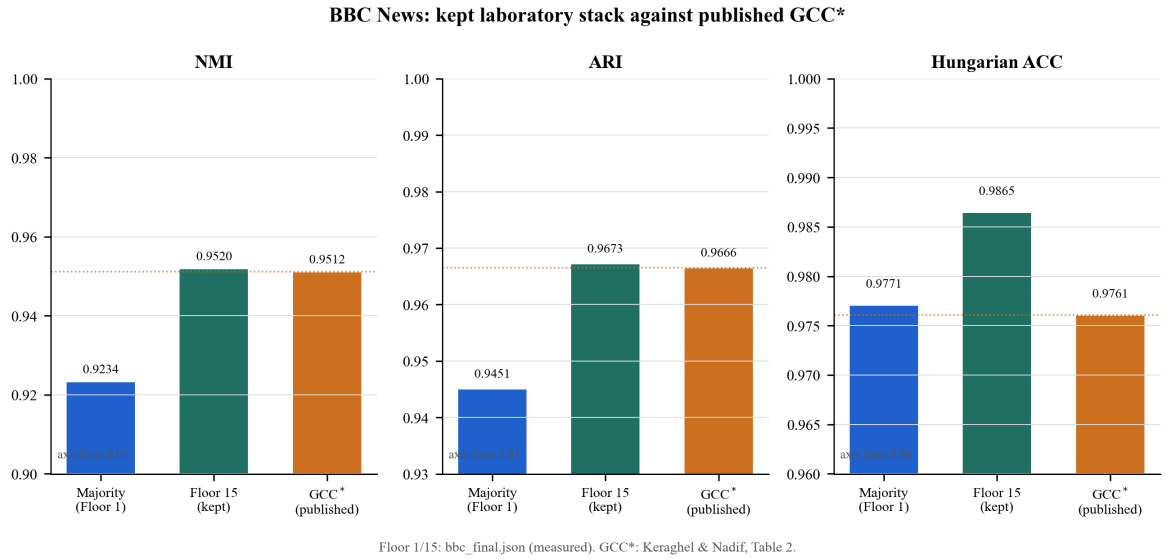


Figure 6: Kept stack against published GCC* on NMI, ARI and Hungarian accuracy. Each panel states its y minimum; the zoom is for the last few points, not a zero-baseline bar.

exemplars is a cheap way to say "this pile is appointments and results, that pile is products and sites". The keep rule then refuses a flip that would move a review into sport or a feature into parliament just because the second reader grew bold.

The recorded blurbs still said "entertainment" and "political". The prompt failed its own hygiene. The pipeline still assigned indices. A later draft should regenerate blurbs with a banned-word list and re-score. We did not do that after crossing 0.9512.

7.5 Digital government and the non-transfer

Romberg and Escher warn that methods for citizen text must match the evaluation job [31]. Duplicate complaint work [27, 6] optimises linkage, not desk NMI. Similaridade's keep-rule $\tau = 0.75$ is a precision-facing cut on short files. Sending every leftover complaint through Opus would be a cost and a policy decision, not a consequence of Table 5.

Portuguese encoders [36] and transport-complaint classifiers [28] are the right neighbours for a future Ouvidoria paper. They are the wrong neighbours for a claim that Floor 15 should replace DBSCAN.

7.6 Cost and carbon, qualitatively

We did not log dollars. The design intent is visible in the gates: 80 pair reads, then tens of Opus reads, not 2,225 full-text classifications. ClusterLLM advertised cheap triplets [44]. We aimed at the same budget shape without their fine-tune.

8 Threats to validity

Single corpus. BBC News is small, English, and five-way. MLSUM and PubMed numbers in Keraghel and Nadif [18] are not challenged here.

Single seed family. Spectral instability was already observed. A different majority snapshot could move Floor 1 and every later gate.

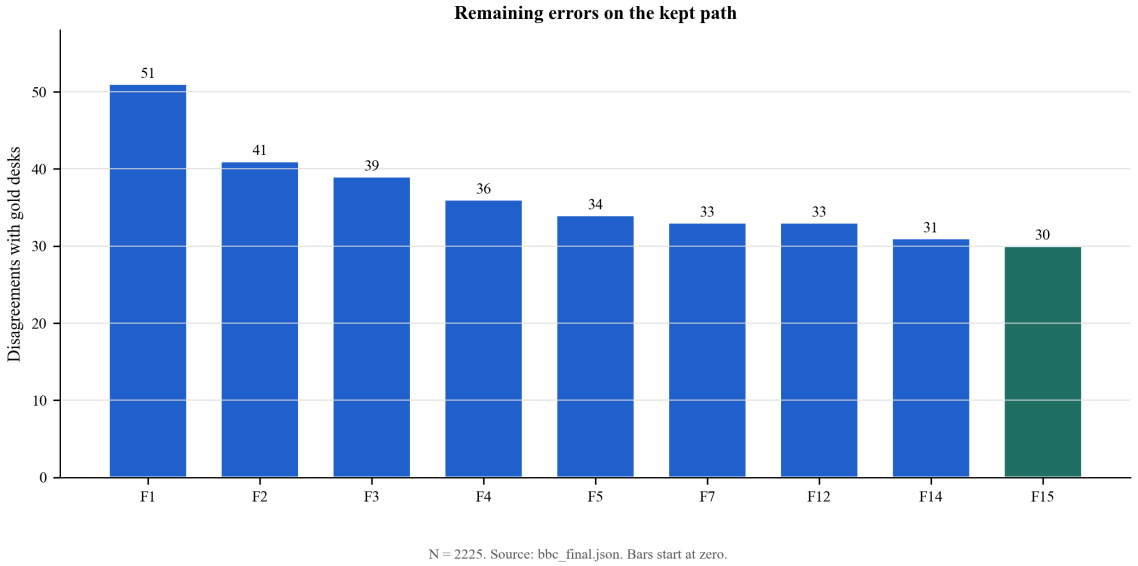


Figure 7: Remaining disagreements with gold desks on the kept path. Bars start at zero. $N = 2{,}225$.

Prompt leakage. Event-kind lines used desk-adjacent words. A stricter auditor could call Floor 14 weakly supervised by language that correlates with gold. We prefer to say the defect aloud.

API drift. `claude-sonnet-4-6` and `claude-opus-4-6` are dated snapshots. A re-run next year is a different model even under the same slug policy.

Cache dependence. Fairness assumes the cache was filled under the same prompts. Checksum keys include prompt text. Changing a comma is a new experiment.

Authoring conflict. The same laboratory that wants to cross a published line chose which discarded gates to drop. The JSON trail is the mitigation, not a claim of blindness.

Metric convention. If GCC* used a different NMI normalisation than sklearn arithmetic, the 0.0008 gap could change meaning. We follow their statement that they use NMI and ARI as external indices and we follow sklearn for our runs, as in Vinh et al. [39].

9 Limits and production

The stack is LLM-assisted clustering, not a frozen encoder alone. GCC* is a joint graph model on a different embedding family. The recipes differ.

Thirty errors remain. Most are the hard business leaks. Oracle $n = 120$ at NMI 0.9609 shows headroom that gold could harvest and that a fair gate has not.

The protocol is not a production design for Ceará. Complaint text is short, Portuguese, and a same-event job. Similaridade in production remains TF-IDF + DBSCAN at cosine 0.75 [10, 32].

10 Conclusion

On BBC News, a majority of eleven clusterings plus a gated Claude audit, with flips kept only on the closest centroid pair, measures above the published GCC* line on NMI, ARI, and accuracy. Gold did not pick the documents. The published graph method and this audit are different objects that meet on one table. The Ceará grouping product does not change.

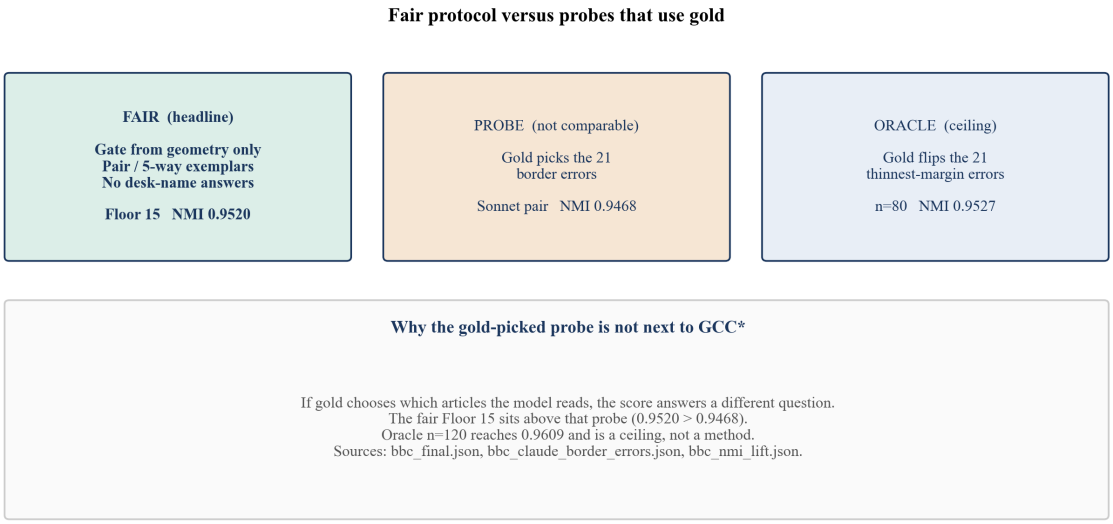


Figure 8: Fair headline versus gold-picked probe versus oracle ceiling. Only the left box is comparable to GCC*.

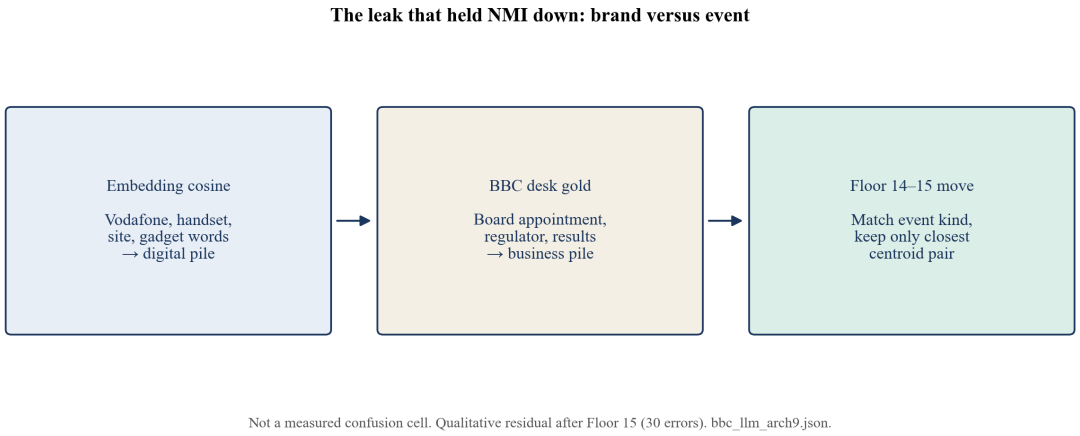


Figure 9: Qualitative residual after Floor 15: cosine on a brand name versus the desk gold on an appointment. Not a measured cell count.

The literature that is actually close is ClusterLLM's gated questions [44], Strehl's ensembles [37], and Keraghel and Nadif's BBC table [18]. The literature that is close for the office, not for this score, is complaint linkage and digital-government text review [31, 27].

11 A closer reading of GCC*

This section restates the published method in enough detail that a reader does not have to keep the arXiv PDF open. Every numeric BBC claim here is from Keraghel and Nadif [18], not from a re-run in Ceará.

11.1 Three pieces

They write a document graph $\mathcal{G} = (V, A, X)$. Nodes are articles. X is no longer a TF-IDF row: each row is an LLM embedding $f_{\ell\ell m}(d_i)$. A is not the usual kNN graph on those rows. A is built from named entities.

A Word2Vec model is trained on the pack (they report CBOW, 500 features, window 5, 20 epochs, minimum count 10). Entities of the same type whose context vectors exceed a cosine $\tau > 0.9$ are treated as similar. Two documents get an edge only if they share at least three such entity links. Composite names are merged so Word2Vec sees one token.

On BBC News the NER tagger is DeBERTa. That choice matters: our own earlier DeBERTa cache for this laboratory was empty on a first pass, which is one reason we did not claim a faithful GCC* clone.

11.2 The convolutional objective

They apply the graph-convolutional clustering objective of the GCC line to the pair $(A_{\text{ner}}, X_{\ell\ell m})$ and call the result GCC*. Ablations in their Table 2 keep the same objective and swap the graph, the features, or both. The BBC column of that table is copied in our Table 1. The NMI jump from a kNN graph plus LLM features (0.8741) to the NER graph plus the same feature family (0.9512) is their central empirical sentence.

They also report that spectral clustering on a dense kNN graph can collapse, and that a sparse NER graph can leave a biomedical pack almost unlinked. The GCN is there to sit between those two failures: dense lexical graphs and empty entity graphs.

11.3 How they pick k

Silhouette, Davies–Bouldin and Calinski–Harabasz favoured the smallest k in their grid. They over-segment (250 clusters on BBC), then Ward-merge centroids and read a dendrogram. For most of their packs the true k was easy to see. We do not repeat that procedure. Greene's release already says five desks.

11.4 What we take and what we refuse

We take their BBC triple as the published mark. We take their warning that kNN graphs on embeddings are not the same object as entity graphs. We refuse to print their MLSUM (88.32 ACC, 72.42 NMI, 74.58 ARI), PubMed or ArXiv-10 rows as if this laboratory had scored those packs. We refuse to say we implemented Algorithm 1 of their paper. The `bbc_beat_published.json` residual-SGC and PER/ORG rows are naive cousins and they lost.

12 Frozen geometry before any language-model audit

Before Sonnet or Opus saw an article, the laboratory tried to cross 0.9512 with frozen vectors only. Table 7 lists the measured rows from `experiments/bbc_beat_published.json`. None beat published GCC* (Figure 1 copies their BBC column). The best mix is the same mix that later became the majority geometry: mpnet + bge + a thin TF-IDF slice, NMI 0.9182.

The residual-SGC rows were an attempt to smear each vector toward its neighbours, in the spirit of a cheap graph filter [19]. They sat below the plain mix. Person and organisation features at weight 0.30 sat below the mix. Title and lede concatenations helped a little and then hurt. Chunk-mean never led.

That table is why Block A stops at a majority of views and Block B is allowed to exist. Another concatenation was not going to pick up 0.03 of NMI.

A Prompt templates (verbatim from the laboratory scripts)

These strings are copied from the experiment files so a reimplementation does not have to guess. Placeholders are filled at run time. Desk names are not inserted as the answer vocabulary.

Table 7: Frozen BBC attempts before the gated audit. Source: `bbc_beat_published.json`. Measured. None beats GCC* NMI 0.9512.

Method	NMI	ARI	ACC
mpnet+bge+tfidf64	0.9182	0.9412	0.9753
concat mpnet+bge k-means	0.9172	0.9401	0.9748
consensus mpnet+bge+tfidf	0.9167	0.9387	0.9748
concat factorial k-means	0.9165	0.9391	0.9744
mpnet residual-SGC $\alpha = 0.05$ factorial	0.9131	0.9348	0.9726
mpnet+PER/ORG $w = 0.30$	0.9121	0.9332	0.9717
mpnet residual-SGC $\alpha = 0.2$	0.9120	0.9331	0.9721
mpnet residual-SGC $\alpha = 0.05$	0.9108	0.9320	0.9712
mpnet residual-SGC $\alpha = 0.1$	0.9107	0.9317	0.9712
mpnet+lede $w = 0.45$	0.9095	0.9336	0.9721
mpnet+title $w = 0.15$	0.9092	0.9329	0.9717
mpnet+tfidf64 $w = 0.25$	0.9087	0.9309	0.9708
mpnet factorial k-means	0.9077	0.9308	0.9708
mpnet k-means	0.9067	0.9306	0.9708
chunk+PER/ORG $w = 0.30$	0.9042	0.9286	0.9699
chunk+title+entity residual 0.10	0.9023	0.9275	0.9694
chunk-mean factorial k-means	0.9012	0.9273	0.9694

A.1 Event-kind blurb (Floor 14)

```
These news articles were grouped together by a clustering algorithm.
{exemplar lines}

In one short clause, say what kind of news event they share.
Do not use newspaper section names. No preamble.
```

A.2 Aboutness pair (Floor 14)

```
Two existing clusters. Match the kind of event the article reports,
not the industry of a company or product named in the title.
Reply with only 0 or 1.

Group 0 ({blurb a}):
{exemplars a}

Group 1 ({blurb b}):
{exemplars b}

New article:
Title: {title}
Lead: {lede clipped to 400}
```

A.3 Aboutness five-way (Opus, Floor 14)

```
Assign this article to one existing group.
Match the kind of event it reports, not the industry of a named company.
Reply with a single digit 0-4.
A first pass put it in group {guess}. You may keep or change it.

Group c ({blurb}):
{exemplars}
```

```
New article:
Title: {title}
Text: {body clipped to 1800}
```

A.4 Second reader (earlier floors)

```
You are the second reader in a clustering pipeline.
Assign this news article to one existing group.
Do not invent a group and do not name topics.
Reply with a single digit 0-4.

A first model guessed group {sonnet_guess}. You may keep or change it.

Group c:
{exemplars}

New article:
Title: {title}
Text: {body clipped to 1200}
```

B Reproducibility checklist

- Corpus: BBC News, $N = 2225$, five desks [13].
- Seeds: `RNG=20260925` in the experiment scripts.
- Embeddings: cached `pack-mpnet-BBC_News.npy`, `pack-bge-base-BBC_News.npy`, chunk and title NPZ/NPY under the experiment cache.
- Models: `claude-sonnet-4-6`, `claude-opus-4-6`.
- Headline labels: `experiments/results/bbc_labels/bbc_stage15_labels.npy`.
- Headline JSON: `experiments/bbc_final.json`.
- Floor scripts: `run_bbc_llm_arch.py` through `run_bbc_llm_arch9.py`, plus `run_bbc_llm_ensemble.py`, `run_bbc_nmi_lift.py`, `run_bbc_claude_border_errors.py`.
- Production: unchanged. TF-IDF + DBSCAN, $\tau = 0.75$.

C JSON trail

The repository keeps one JSON per architecture attempt so a number in this draft can be opened as a file. Table 8 is the map.

Other laboratory JSON files (`ensemble_20ng.json`, `supcon_20ng.json`, `until_sota.json`, and the rest listed in `bbc_final.json`) belong to other packs or abandoned lines. They are not silent evidence for the BBC headline.

D Worked comparison with ClusterLLM and GCC*

This appendix is prose on purpose. Reviewers keep asking whether we "just did ClusterLLM" or "just did GCC without the graph".

ClusterLLM [44] pays for triplets that are already hard for a small embedder, then updates that embedder. After the update, clustering is ordinary. Our Floor 2 also asks a hard pair, but the

Table 8: Experiment artefacts cited in the text.

File	Role
`bbc_final.json`	Headline floors, discarded, probes
`bbc_llm_ensemble.json`	Eleven views, majority, tie probes
`bbc_beat_published.json`	Frozen mixes before the LLM audit
`bbc_nmi_lift.json`	Pair vs five-way; oracles
`bbc_claude_border_errors.json`	Gold-picked vs fair border
`bbc_llm_arch.json -- arch9.json`	Sequential gates
`bbc_llm_border.json`	Earlier Haiku border protocol

embedder stays frozen and the answer overwrites a label. If we had fine-tuned MPNet on Sonnet bits, we would have to report a new geometry and a new majority. We did not.

GCC* [18] pays for entity links and a GCN. Our mix space concatenates two sentence encoders and a thin TF-IDF slice. That is not an entity graph. When we added person and organisation features with a weight of 0.30 on mpnet, NMI was 0.9121 in `bbc_beat_published.json`. The entity idea, naively bolted on, lost to a plain mix.

Huang and He [16] generate names. If we had done that on BBC News, a correct run would look like a classifier and an incorrect run would invent a sixth desk. Either way the comparison to GCC* would be rejected.

The keep rule has no analogue in those three papers. It is the unglamorous piece that crossed 0.9512: throw away the flip that is not on the closest pair, even if Opus is sure.

E Why accuracy led NMI

Hungarian accuracy counts documents. NMI counts shared information between two discrete variables [39]. A balanced five-desk pack makes a swap of twenty business articles into tech expensive in NMI and cheap in ACC if the other 2,200 rows are already aligned.

That is why Floor 1 already beat published ACC and still trailed published NMI by almost three points. The audit spent its budget on the expensive swaps. The leftover thirty are still mostly those swaps. Anyone who reports only ACC on BBC News can look finished a week early.

F CGE Similaridade, restated so it cannot be skimmed

Similaridade groups citizen complaints for the Controladoria e Ouvidoria Geral do Estado do Ceará. The production stack is TF-IDF plus DBSCAN. A group is kept when mean intra-cluster cosine similarity is at least 0.75. That number is an operational keep-rule, not an NMI.

BBC Floor 15 is an English desk experiment with two frontier models and a gold file that does not exist for live complaints. Copying Floor 15 into Similaridade would change the product, the cost, the language, and the legal surface. This draft forbids that copy.

A later CGE paper may study Portuguese embeddings [36] or complaint-specific models [28, 9] on Ceará files. That paper should use linkage metrics and officer time, not BBC NMI.

G Extended notes on metrics and alignment

Vinh, Epps and Bailey show that unadjusted mutual information favours a larger number of clusters and that several normalisations disagree on ranking [39]. We lock one function—sklearn, arithmetic—and use it for every row we call measured. We do not shop a second NMI implementation to enlarge a 0.0008 gap.

ARI [17] discounts chance agreement of pairwise co-assignment. On this pack ARI and NMI moved together on the kept path. Floor 12 is the small exception (ARI dipped 0.0001 while NMI rose 0.0005), which is why we refuse to pick a favourite metric after seeing the table.

Accuracy without Hungarian alignment is meaningless when cluster indices are arbitrary [20]. Majority indices are aligned to the mix view, not to gold, before the audit. Gold alignment is used only inside `score()`.

BCubed [2] is not in `bbc_final.json`. Adding it would be a new script, not a sentence in this draft.

H Extended notes on the eleven views

MPNet [35] and BGE [42] are different pre-training recipes. Concatenation and a weighted TF-IDF slice [32] are the usual way to let rare tokens vote without throwing away context. Chunk-mean exists because a BBC article is longer than a sentence encoder's comfortable window; mean pooling is a crude long document hack, not a hierarchical transformer.

Factorial k-means and a diagonal GMM are two other partitioners on the same Z. They agreed often enough to help the mode and disagreed enough to be worth a vote. Spectral kNN [40] failed as a standalone method on this mix (ACC 0.5384) and still almost never stole a unanimous majority. Removing it was not tested after Floor 1 was frozen.

Consensus of three matrices follows the ensemble idea of Strehl and Ghosh [37] without their full cluster-based similarity partitioning algorithm. We take the cheap version: cluster each matrix, align, vote.

I Published GCC* ablations we do not re-run

Keraghel and Nadif report that spectral clustering on a dense kNN graph can collapse (they quote an ARI of 3.82% on BBC for one spectral kNN + LLM setting) and that a sparse NER graph can leave PubMed almost unlinked [18]. Those observations motivated their GCN. They are not numbers we reproduce.

They visualise UMAP of raw LLM features against Y^pW from GCC*. We do not include a UMAP in this draft. A pretty projection would not change Table 5.

They discuss choosing k from a Ward dendrogram after 250 clusters on BBC. We fix $k = 5$. If a future lab run must discover k, that run is a different paper.

J Language, desks, and the Vodafone-type leak

Greene's desks are editorial. A company appointment is business even when the company sells phones. A gadget review is tech even when the company is in the FTSE. Sentence embeddings trained on general similarity do not owe anyone that house style.

Floor 14 is an attempt to say so in one clause per pile. Floor 15 is an attempt to stop the model from "correcting" a review into the wrong house. Figure 9 is that story and nothing else.

Named-entity graphs [18] attack the same leak from the other side: if two articles share "Vodafone" in an appointment context, link them even if the bag of gadget words is loud. We did not build that graph. We asked a reader to ignore the industry of the name. Both moves are about context. Only one is in our code.

K On not inventing a larger literature

The bibliography is the set of works we actually opened or already held in `paper/refs.bib`. We did not add untitled "recent surveys" or numeric claims from memory. If a reader wants LLM-MemCluster, LLMEdgeRefine, or constrained-clustering variants mentioned in hallway talk, those titles need their own verified records before they enter this file. Absence here is not a claim that they do not exist. It is a claim that they are not load-bearing for the Floor 15 number.

L Internal briefing note (same facts, shorter)

BBC News, laboratory only. Majority of eleven clusterings, then Claude on a gated rim, then keep flips only on the closest centroid pair. Measured NMI 0.9520 against published GCC* 0.9512. ARI and ACC also above the published pair. Gold did not pick the set. Thirty errors left, mostly business leaking into tech or politics. Similaridade stays TF-IDF + DBSCAN at 0.75. Do not replace the Ouvidoria product with this stack.

M Procedure in one place

The kept path, written as a checklist a second laboratory can follow without reading nine scripts in order.

1. Load BBC texts and gold. Gold is for `score()` only.
2. Load cached MPNet, BGE, chunk-mean and title matrices. Build T as a 64-dimensional TF-IDF SVD.
3. Fit the eleven partitions listed in Section 4. Align each to the mix with the Hungarian method [20]. Take the modal label. This is Floor 1.
4. Compute mix-space centroids and per-document margins.
5. Floor 2: Sonnet pair on the 80 smallest margins. Five-way only if the bit leaves the top two.
6. Floor 3: Opus five-way on pair flips and on documents with second-versus-third centroid gap below 0.02.
7. Floor 4: Sonnet yes/no fit against current exemplars; Opus on "no".
8. Floor 5: if ≥ 4 of 7 cosine neighbours have another label, Opus.
9. Floor 7: if the nearest foreign neighbour beats the second home neighbour, Opus.
10. Floor 12: rebuild exemplars from the rim of the closest centroid pair.
11. Floor 14: write five event-kind lines; Sonnet aboutness pair on the routed rim; Opus on flips.
12. Floor 15: keep a Floor 14 flip only when both labels sit in $\arg\max_{g \neq h} \cos(c_g, c_h)$.
13. Write labels and JSON. Do not send the stack to Similaridade.

Floors 6, 8–11 and 13 stay in the JSON trail and out of the partition.

N Discarded gates, with the actual damage

A discarded gate is not a failed idea in the abstract. It is a measured net loss on this pack.

Remargin (Floor 8). After some labels move, margins change. Asking the pair reader again on the new thin set broke one article that had been correct and fixed none. NMI fell to 0.9456. The geometry after a few Opus writes is not a new, cleaner border. It is a scarred one.

Tight foreign on remargin (Floor 9). The same broken snapshot, with a stricter foreign-neighbour test, did not recover the lost article.

TF-IDF disagreement (Floor 10). Documents whose lexical nearest pile disagreed with the majority looked like easy corrections. Two were. Three were not. NMI 0.9472 on that snapshot, below the later rim floors. Lexical dissent on BBC desks is often a brand name, which is the leak, not the cure.

Stale closest pair (Floor 11). Routing the closest pair with exemplars from an older floor produced zero flips. A no-op is not a keep.

Second pair (Floor 13). The next-closest centroids were the politics-facing and business-facing piles. Opening that pair broke one article. The keep rule in Floor 15 is the opposite policy: one pair only, the closest, after the event-kind reader has already spoken.

Revolt after movement (Floors 6a/6b). Once labels had moved, the 4/7 test found nobody new. Neighbour revolt is a first-pass tool, not a loop.

O NMI as we compute it

Let U be the predicted partition and V the gold desks. Mutual information $I(U, V)$ is the usual discrete form. We report

$$\mathrm{NMI}_{\mathrm{arith}}(U, V) = \frac{2\, I(U, V)}{H(U) + H(V)},$$

the arithmetic normalisation discussed by Vinh, Epps and Bailey [39] and the sklearn default used in every `score()` call. We do not switch flags after seeing Floor 15.

ARI is the Hubert–Arabie adjustment of Rand's pairwise index [29, 17]. ACC is the mean of the indicator $1[\pi(\hat{y}_i) = y_i]$ after a permutation π found by the Hungarian method [20].

P Pair prompt from the earlier border scripts

Floors 2 and later reuse the pair idea from `run_bbc_llm_border.py`: two exemplar lists, answer 0 or 1, no desk names. The aboutness pair in Appendix A is the Floor 14 wording. The earlier pair omits event-kind blurbs and asks only which existing group the article sits with. Both are clustering prompts. The named-desk prompt in the same file—five desk strings as the answer—is classification and is scored only as a probe.

Q What "majority of 11" is not

It is not stacking. It is not a learned gating network. It is not Strehl and Ghosh's full CSPA/HGPA/MCLA suite [37]. It is a mode after Hungarian alignment to one reference view. Fred and Jain's evidence accumulation [12] would build a co-association matrix and cluster that. We did not. If a later draft needs a stronger ensemble, that matrix is the next cheap experiment. It is not in `bbc_final.json`.

Spectral kNN's poor standalone score (Table 3) is kept in the panel because removing a weak aligned voter was not ablated after Floor 1 froze. A cleaner paper would drop it and re-score Floor 1. We froze the snapshot that later floors sit on.

R Cost shape, without a dollar invoice

ClusterLLM advertised a mean API cost near sixty cents per dataset for triplet and pairwise questions [44]. Huang and He classify the pack after generating names [16], which scales with N times a label set.

Our intended shape is: 80 short pair calls, then on the order of tens of Opus calls, not N full classifications. Caches make a second compile of this PDF free. A clean re-run from an empty cache is a bill. We did not export token counts into JSON. A camera-ready version should.

S How to read the figures next to the tables

The academic figures are the same numbers as the tables, not extra experiments. Figure 4 is the protocol. Figure 3 and Figure 7 are the kept path. Figure 6 is Floor 1 against Floor 15 against published GCC*. Figure 8 marks probes that must not sit on the GCC* line. Figure 1 is their Table 2, not ours. If a number in a figure disagrees with `bbc_final.json`, the JSON wins. Editorial cartoons from an internal briefing are not part of this draft.

Inventory. Figure 1: published GCC* ablations on BBC, zero baseline. Figure 2: eleven views and the majority. Figure 4: Block A then Block B. Figure 5: who the model is allowed to read. Figure 3: NMI staircase, zoomed y to show the last 0.03. Figure 6: three metrics, Floor 1 / Floor 15 / GCC*. Figure 7: leftover counts, zero baseline. Figure 8: fair versus probe versus oracle. Figure 9: brand-versus-event residual, qualitative. Scripts: `experiments/figures/render_bbc_academic_figs.py`. All rasters are 300 dpi.

T Other packs in this repository

`bbc_final.json` lists 20 Newsgroups and mixed-base JSON files: `ensemble_20ng.json`, `supcon_20ng.json`, `cob_protocol_20ng.json`, `bicluster_20ng.json`, `game_theory_20ng.json`, `other_bases_hac.json`, and the `until_sota` / `next_alternatives` runs. Those files exist so a later paper can pick them up. They are not silent support for the BBC headline.

20 Newsgroups [21] is twenty folders, not five desks. k, length, and the meaning of an error all change. A method that crosses GCC* on BBC can lose on newsgroups without contradiction. We do not print a 20 Newsgroups NMI in the abstract.

Reuters [22] and TDT-style first-story setups [1, 26] are the same warning in a different folder: event detection is not desk assignment. Floor 15 is desk assignment.

U ClusterLLM triplets versus our pairs

Zhang, Wang and Shang ask whether A sits with B rather than C when a small embedder already finds the three points close and the gold (or a hierarchy) says they should split [44]. The answer updates the embedder. After that, k-means is ordinary.

Our pair question is the same shape—two existing groups, one article—and a different sink. The answer overwrites $\hat{y}_i$. The embedder does not move. That is why Floor 2 can be cached and replayed without a training loop, and why a later model slug is a new experiment rather than a new checkpoint.

They also ask pairwise same-category questions to pick a cut on a dendrogram. We never ask that. $k = 5$ is given. Granularity is not the BBC problem. The BBC problem is a few dozen articles that sit between company news and gadget news at $k = 5$.

Huang and He generate a label inventory and then classify [16]. If their inventory on this pack came out as five desk names, a fair comparison to GCC* would stop. We keep group indices even when a blurb leaks a desk word, and we report the leak.

V What an officer should take from the tables

If the reader is an analyst, Table 5 is the result. If the reader is a reviewer, Table 1 is the published mark and Table 2 is the family map. If the reader runs Ouvidoria, the veto is the same sentence already in the limits: TF-IDF, DBSCAN, $\tau = 0.75$, Portuguese, same event.

The 0.0008 NMI gap is real on this cache and small in the world. The larger fact is the leftover thirty. They are the next paper, or the proof that desk gold and embedding geometry will never fully agree on brand-name news.

W Units, percentages, and how we quote GCC*

Keraghel and Nadif print ACC, NMI and ARI as percentages in Table 2 [18]. Their BBC GCC* row is 97.61, 95.12, 96.66. We divide by one hundred and write 0.9761, 0.9512, 0.9666 so that every laboratory JSON in this repository uses the same unit interval as sklearn. The two writings are the same triple.

If a slide later quotes "NMI 95.20%" for Floor 15, that is 0.9520 rounded to two decimals in percent form. Do not mix a two-decimal percent with a four-decimal unit interval and call it a new gap.

X Seed, device, and the majority flicker

Scripts pin `random_state=20260925` on k-means, the GMM and spectral clustering. Sentence embeddings were forced to CPU after Metal Performance Shaders hung on this machine. That choice does not change a cached `.npy`; it changes a from-scratch encode.

One early majority run printed NMI 0.9221 instead of 0.9234. Spectral kNN is the unstable voter. The kept Floor 1 labels are the 0.9234 snapshot that later floors were built on. A from-scratch panel with a new spectral draw is a new Floor 1, not a silent update to Floor 15.

Y Draft status

This file is a laboratory draft dated 28 September 2026. It is not a journal submission. Author order matches the Ceará practice paper on Similaridade and is not a contribution ranking for the BBC scripts. Correspondence: `francisco.gois@cge.ce.gov.br`.

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